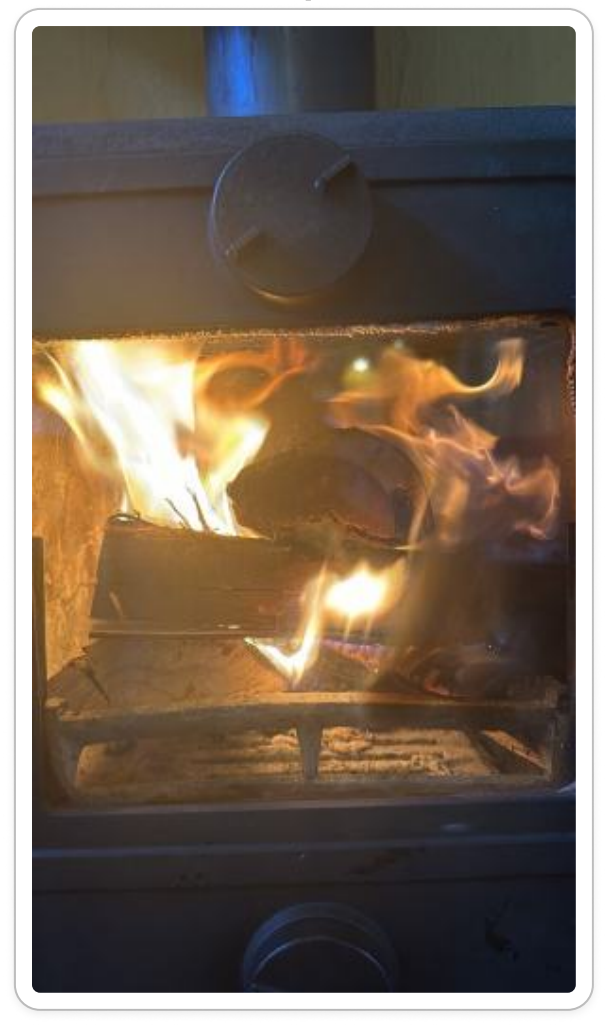
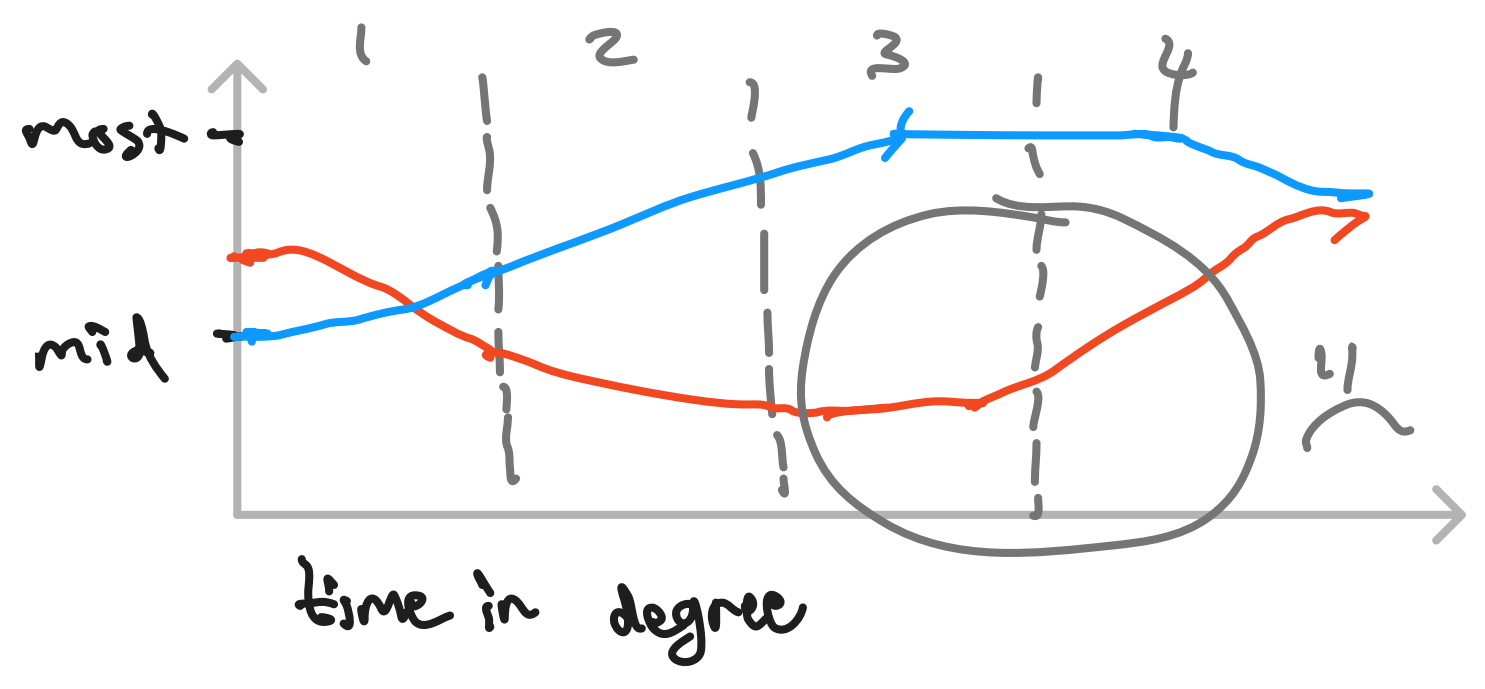
Attendance pic



EEE3094S Module 2 Lecture 1

- engineering-ness
- hardship

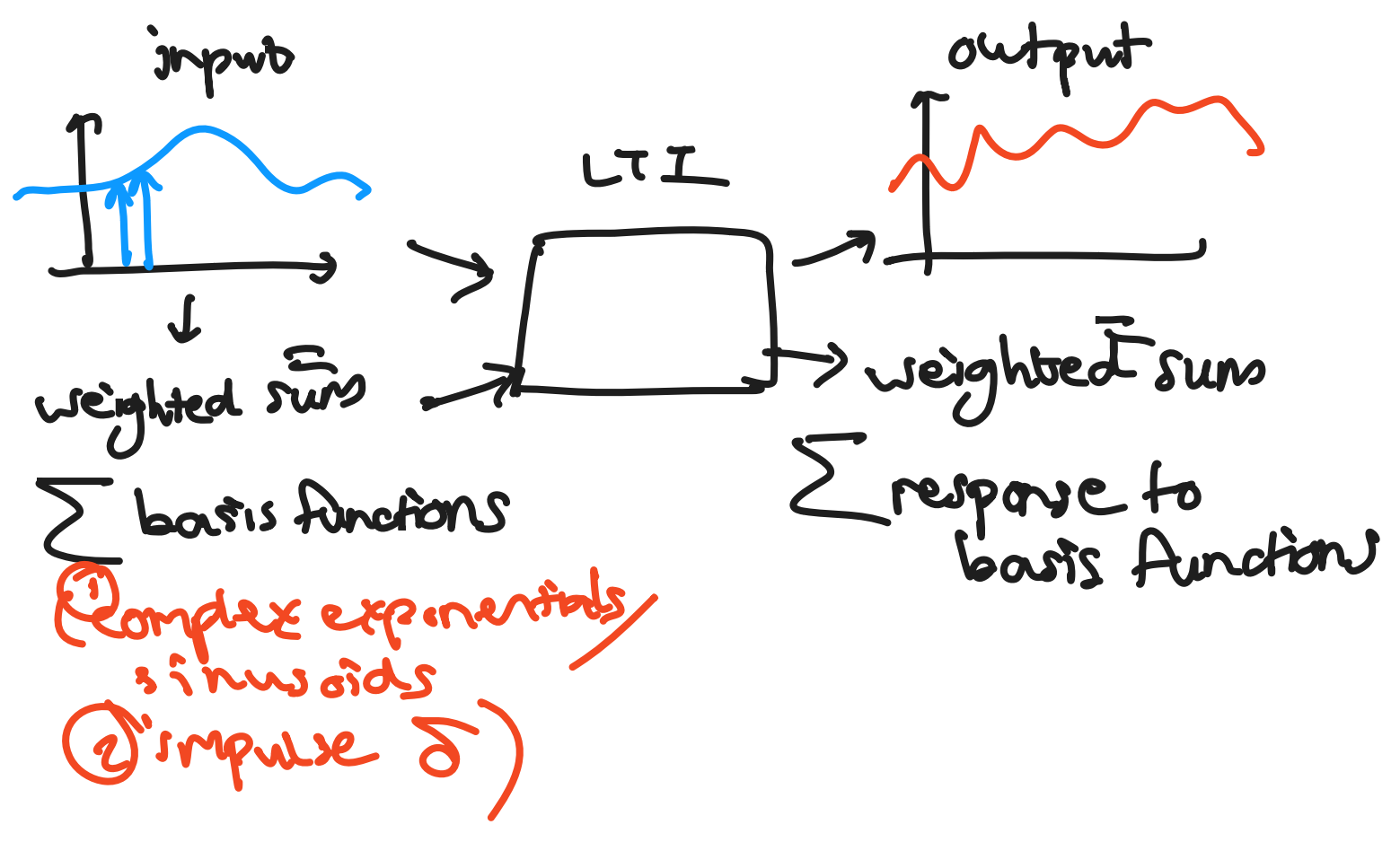


Levels of 3094S:

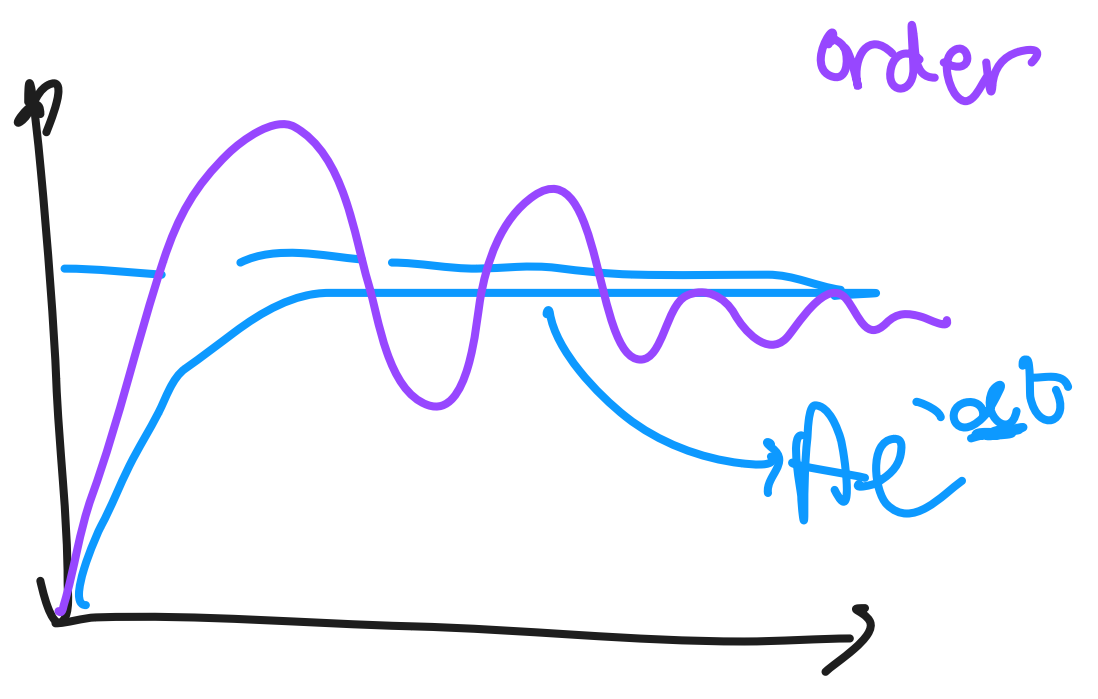
Level 0:
main ideas: linearity + time invariance - basis functions

question: Why do we focus on LTI systems?

recommended watching: https://www.youtube.com/playlist?list=PLZHQObOWTQDPD3MizzM2xVFitgF8hE_ab


input
LTI
output
weighted sums
Σ basis functions
① complex exponentials / sinusoids
② impulse δ
weighted sums
Σ response to basis functions

oscillation = 2nd order

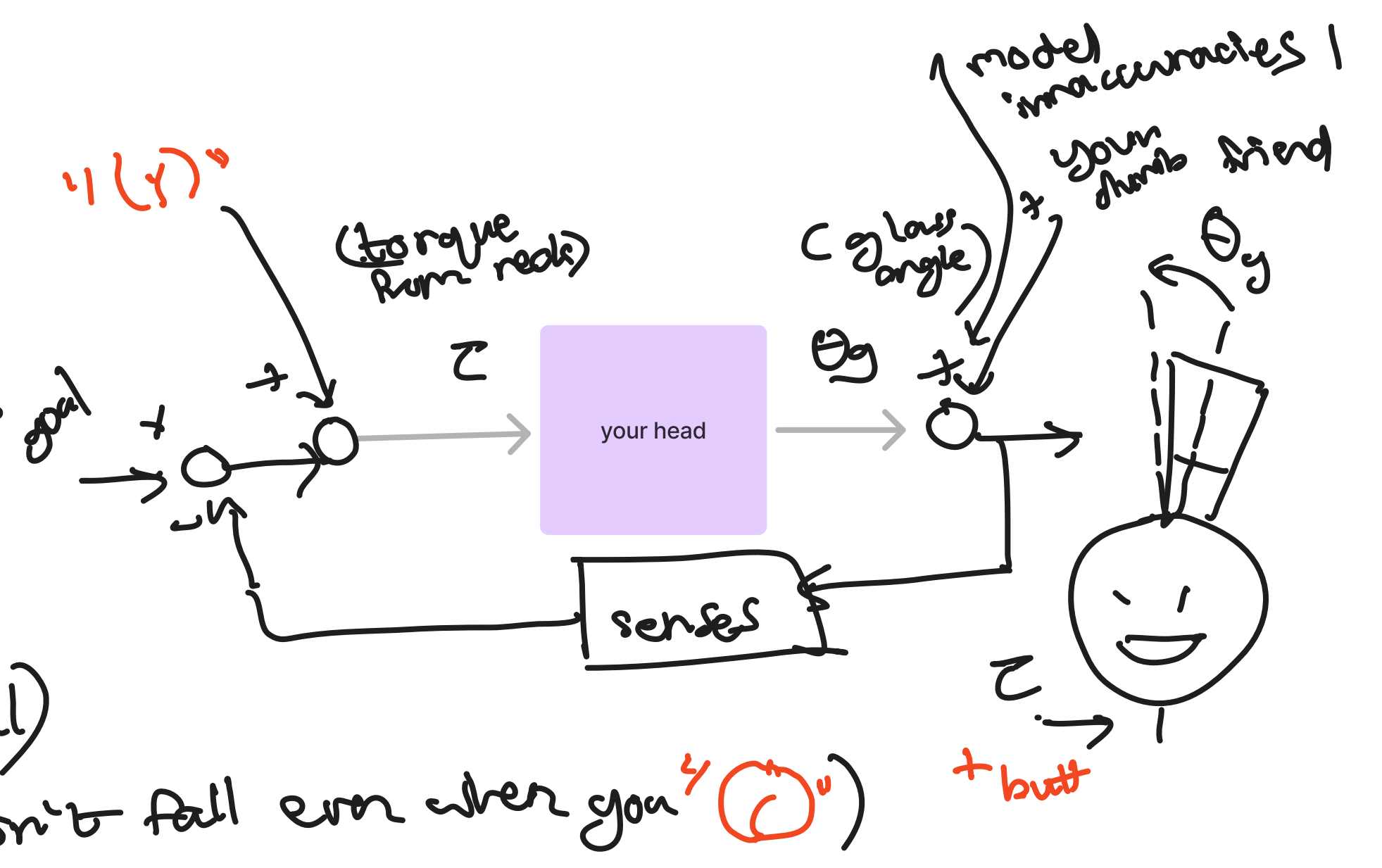


↓
sinusoid
$Ae^{(\sigma+j\omega)t} + Ae^{(\sigma-j\omega)t}$
both need to be present if pole is imaginary

Level 1:
question: How do LTI systems behave?

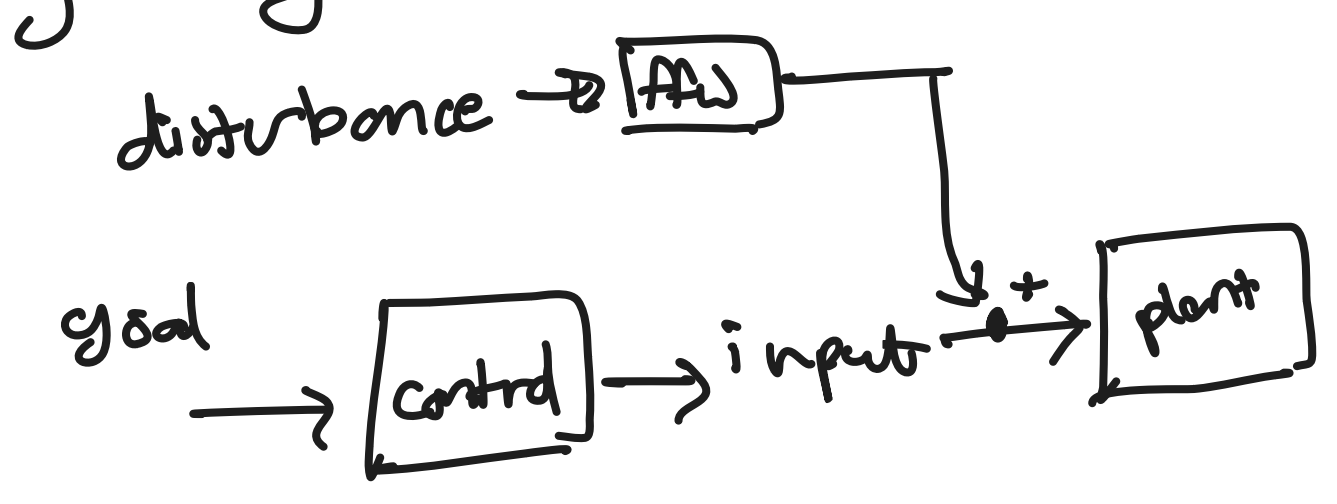
Level 1.5: question - How to we want our systems to behave?

E.g. how to balance a drink on your head while you shake your ass


"1(t)"
goal
+
−
+
(torque from neck)
τ
your head
θg
(glass angle)
+
model inaccuracies / your dumb friend
θg
senses
τ
+ butt

what do we want?

1) Stability (drink doesn't fall)
2) robust to disturbances (doesn't fall even when you "🍑")
3) accuracy (tracking (glass is upright)
4) style / "controlledness" (how 'nicely' do you balance the drink)
= transient response


disturbance
FFW
goal
control
input
+
+
plant

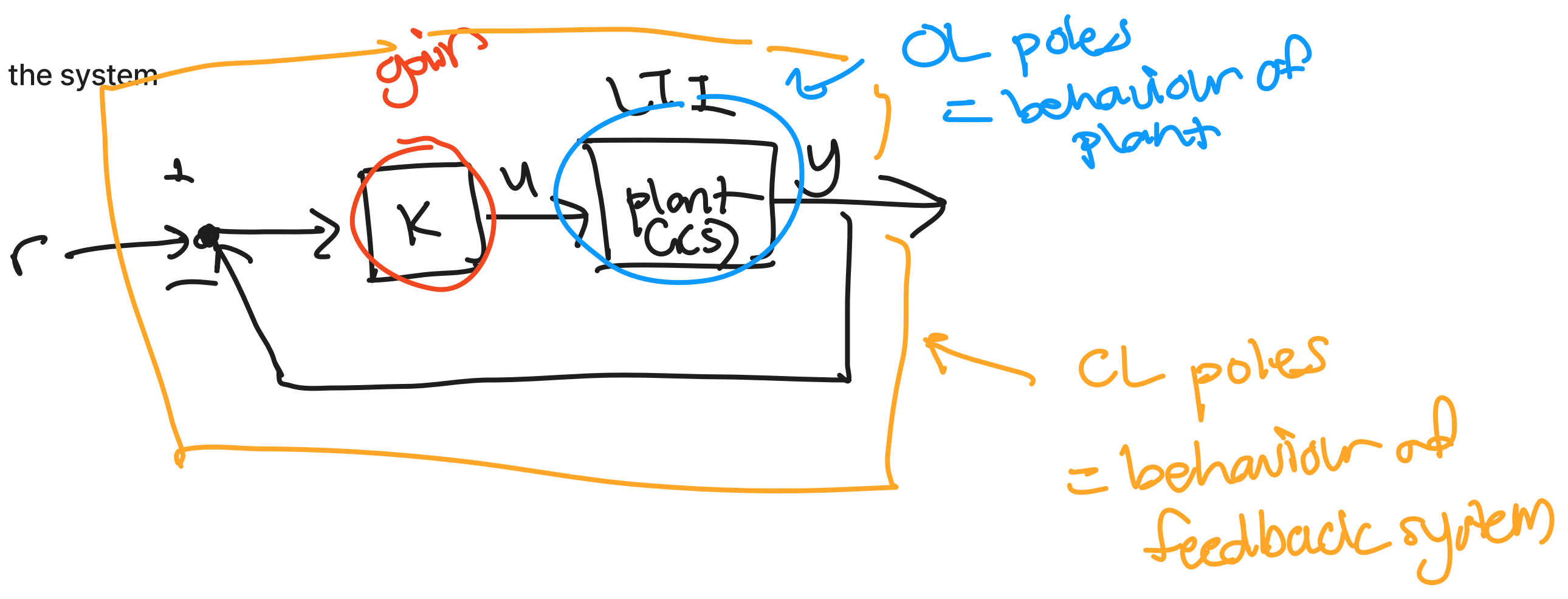
Q2: why feedback? (as opposed to...)

Open loop = input applied with no knowledge of output

feedforward control: preemptive compensation for known disturbance

feedback adds knowledge of the output and allows the system to react to it.

Introducing feedback


gains
LTI
OL poles = behaviour of plant
r
+
−
K
u
plant G(s)
y
CL poles = behaviour of feedback system

Big gain (generally)
→ faster
→ more oscillatory
→ maybe risk of becoming unstable

The maths: Closed Loop Poles

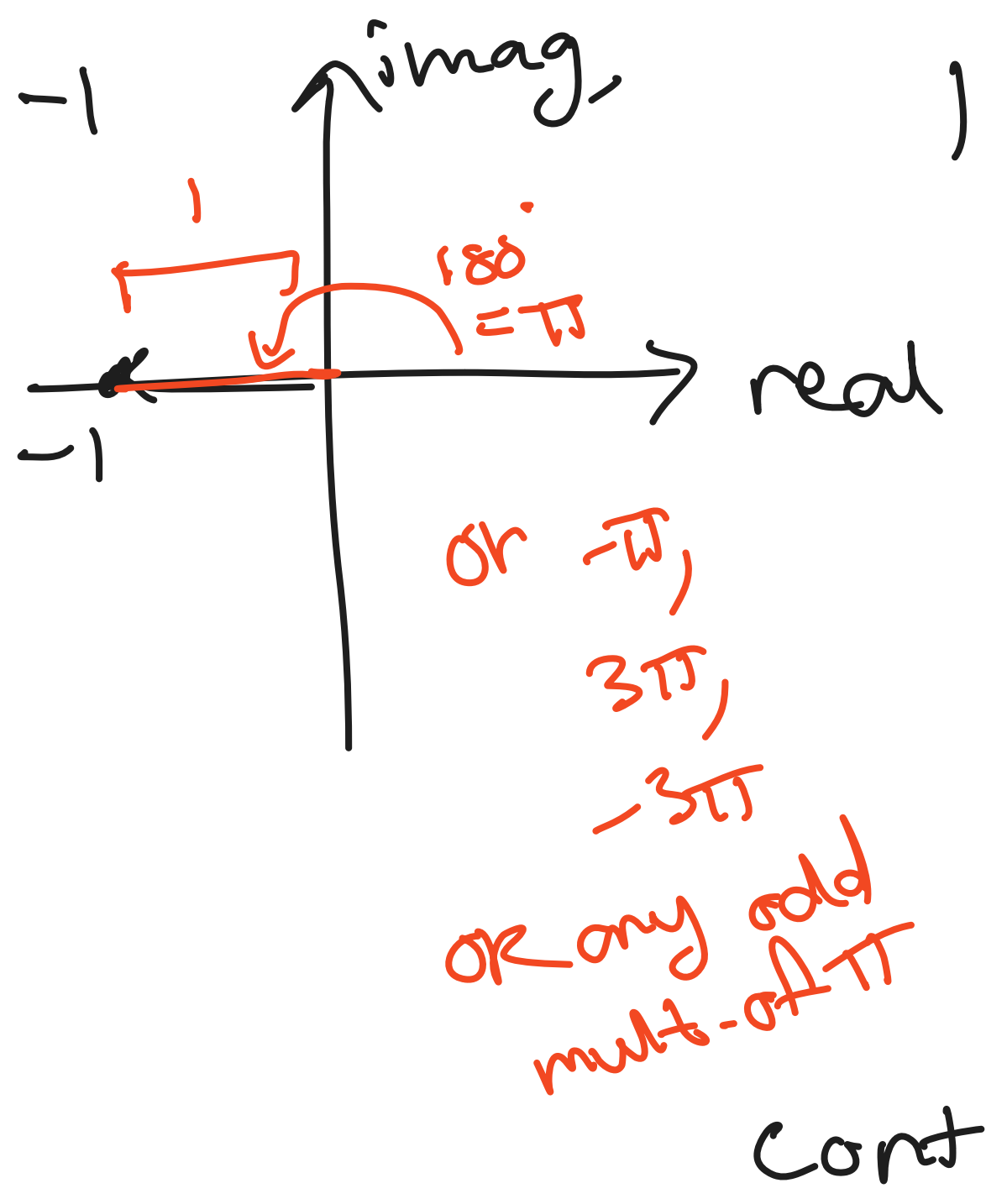
$$G_{cs}(s) = \frac{KG(s)}{\boxed{1+KG(s)}}$$

Closed loop characteristic eq: $1 + K.G(s) = 0$
↳ solutions = CL poles

↳ ∴ $K.G(s) = -1$

Note: $K > 0$

↖ If s makes ↗ true means s is a closed loop pole @ that value of K


−1
imag
1
180° = π
real
−1
or −π, 3π, −3π
or any odd mult. of π

$|K.G(s)| = 1$

$K.|G(s)| = 1$

∴ $|G(s)| = 1/K$

$\angle K = 0$

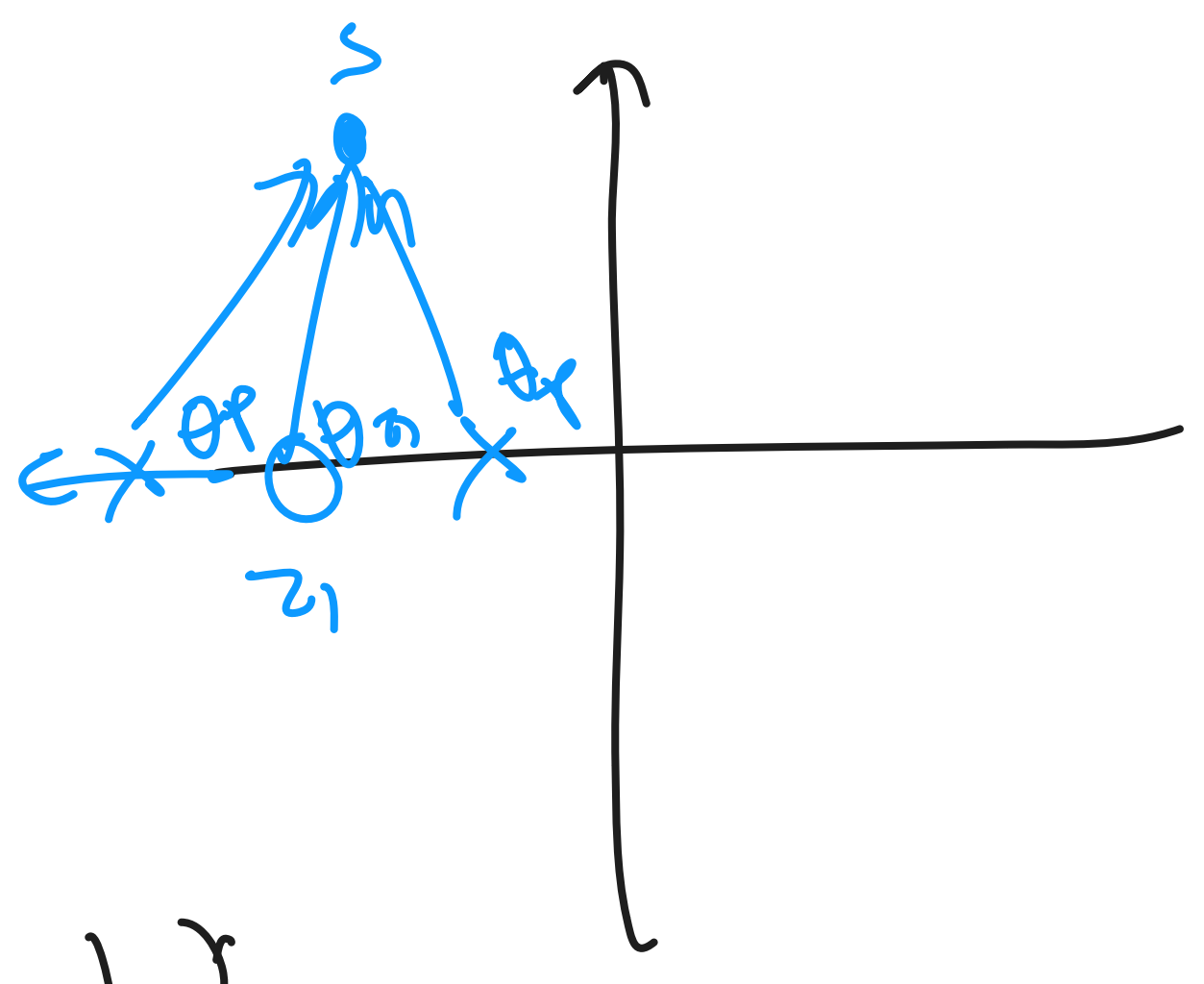
$\angle K.G(s) = \pi + \ldots$
$\angle G(s) = \pi + \ldots$ } angle criterion

Control = choosing CL poles that = desired behavior
1) finding s that satisfy angle crit.
2) finding K that 'activates' chosen pole.

Note: we are designing the CL system using knowledge only of OL.

Satisfying Angle crit

$$G(s) = \frac{N(s)}{D(s)} = \frac{(s-z_1)(s-z_2)\ldots}{(s-p_1)(s-p_2)\ldots}$$



$\angle G(s)$:-

$a_1 e^{b_1 j} \times a_2 e^{b_2 j} = a_1 . a_2 . e^{(b_1+b_2)j}$ angles add

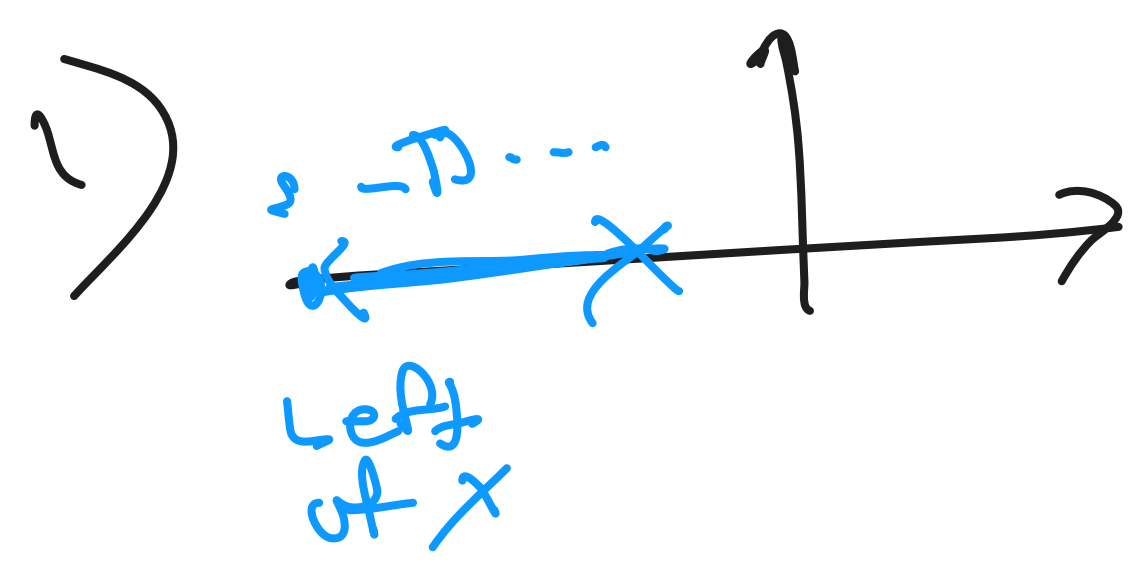
$\angle G(s) = \left(\angle(s-z_1) + \angle(s-z_2) + \ldots\right) - \left(\angle(s-p_1) + \angle(s-p_2) \ldots\right)$

zero angles — pole angles

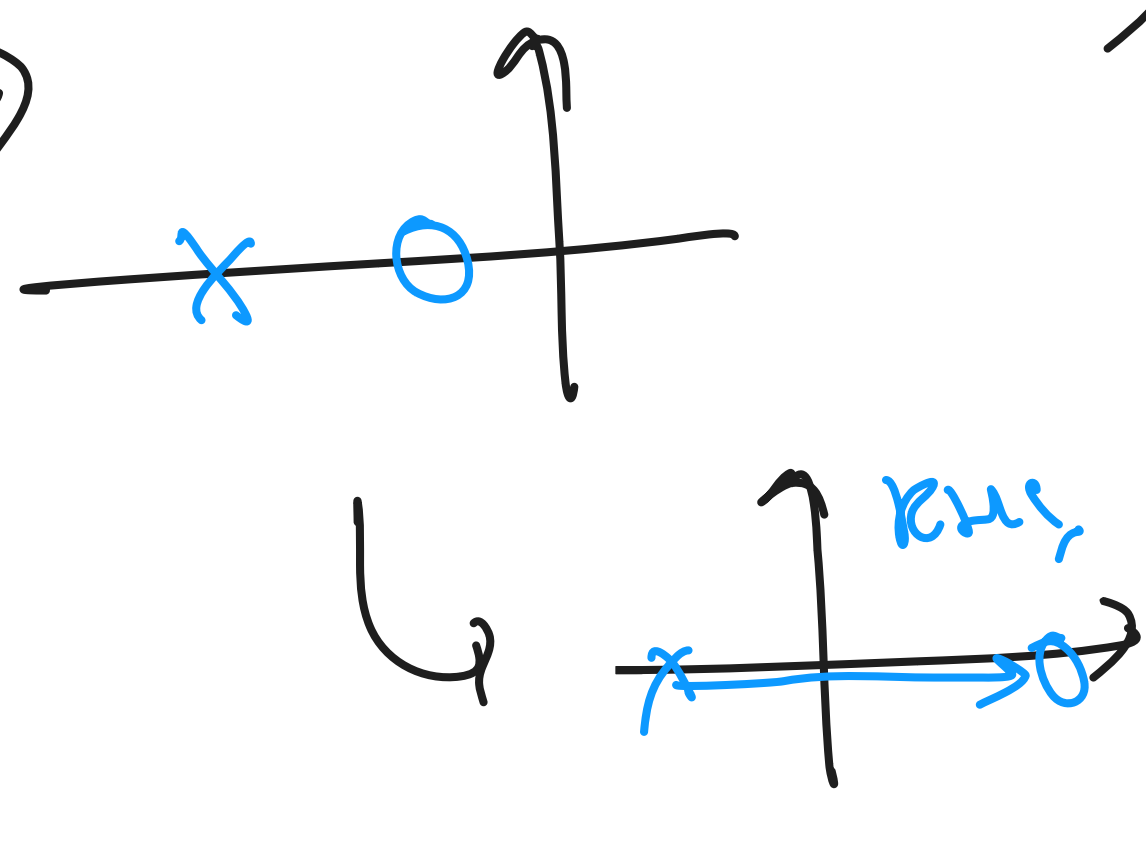
Homework:

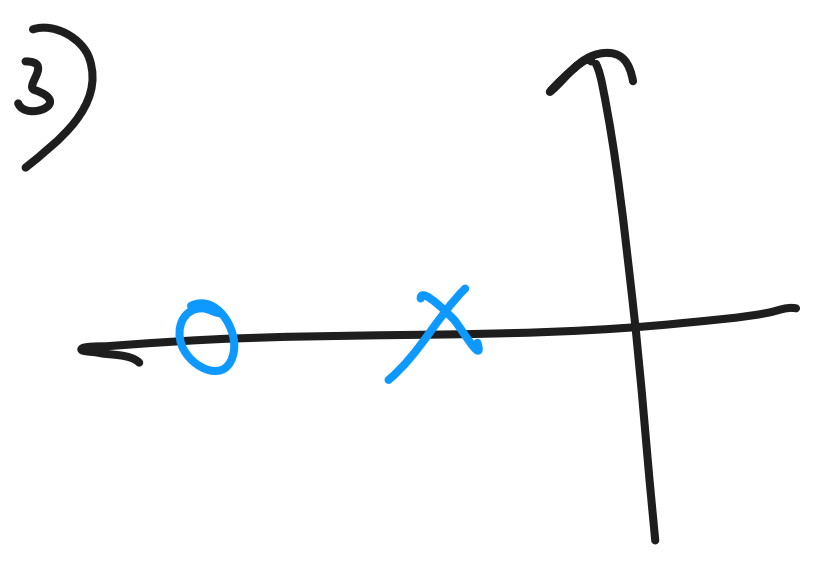
For all possible combos of poles & zeros for 1st & 2nd order systems, find: where can s be so the A.C. is satisfied?

1)



2)



3)